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RESEARCH PAPER

Plant species with larger extrafloral nectaries produce better quality nectar when needed and interact with the best ant partners

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Abstract

Few studies have explored the phenotypic plasticity of nectar production on plant attractiveness to ants. Here, we investigate the role of extrafloral nectary (EFN) size on the productivity of extrafloral nectar in three sympatric legume species. We hypothesized that plant species with larger EFNs (i) have higher induced nectar secretion after herbivory events, and (ii) are more likely to interact with more protective (i.e. dominant) ant partners. We target 90 plants of three *Chamaecrista* species in the field. We estimated EFN size and conducted field experiments to evaluate any differences in nectar traits before and after leaf damage to investigate the phenotypic plasticity of nectar production across species. We conducted multiple censuses of ant species feeding on EFNs over time. Plant species increased nectar descriptors after leaf damage, but in different ways. Supporting our hypothesis, *C. duckeana*, with the largest EFN size, increased all nectar descriptors, with most intense post-herbivory-induced response, taking its place as the most attractive to ants, including dominant species. EFN size variation was an excellent indicator of nectar productivity across species. The higher control over reward production in plants with larger sized EFNs reflects an induction mechanism under damage that reduces costs and increases the potential benefits of indirect biotic defences.

Keywords: Ant–plant interactions, Caatinga, cost, extrafloral nectar, induced defence, mutualisms.

Introduction

Generalized mutualisms involving multiple partner species are ubiquitous in natural systems (Bronstein, 2015), and the frequency and outcome of these interactions may vary considerably over space and time (Billick and Tonkel, 2003; Rudgers *et al.*, 2010; Vilela *et al.*, 2014). This variation is explained by partner traits involved in these interactions (Bronstein, 1994a). Plant traits associated with resource availability to mutualist partners are diverse and determine the

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chances of establishing local interactions (Bronstein, 1994b, 2015). Mutualistic traits have been optimized over evolutionary time to ensure the increased ability of the partners to take advantage of services or cost-saving investments in resources (Connor, 1995; Bronstein, 2015). In this context, knowledge of the ecological and evolutionary processes that structure generalized mutualisms is gained through a comprehensive understanding of the variation in partner traits associated with mutualisms as determinants of interactions between particular species pairs (Bronstein *et al.*, 2006; Leigh, 2010). However, most studies have focused mainly on the variability of mutualism outcomes while neglecting between-partner traits that promote and regulate mutualistic interactions.

Ant–plant interactions are among the most common and most investigated mutualisms, and the attributes mediating these interactions vary widely among partners (Bronstein, 1998; Lanan and Bronstein, 2013; Ojha *et al.*, 2022). In ant–plant protection mutualism, plants provide food and/or shelter to the ants in exchange for partial or permanent protection as ants remove or reduce the activity of herbivores (Rudgers, 2004; Rudgers and Gardener, 2004; Rosumek *et al.*, 2009; Nascimento and Del-Claro, 2010; Leal and Peixoto, 2016). Generalized ant–plant protection mutualism is commonly mediated by extrafloral nectaries (EFNs) that secrete a solution rich in sugars and other solutes, especially on vegetative plant organs such as leaves. These glands provide a feeding source for ants without any barrier restricting their access (Rico-Gray and Oliveira, 2007; Marazzi *et al.*, 2013; Roy *et al.*, 2017). Therefore, all nectar-feeding ants in the local assemblage may interact with these plants. The extrafloral nectar consumed by ants differs significantly between plant species, mainly in terms of volume, sugar concentration, and temporal secretion dynamics (Nicolson *et al.*, 2007). This variation is partly associated with the size and diversity of morphological arrangements of EFNs (Díaz-Castelazo *et al.*, 2005; Marazzi *et al.*, 2019). For example, larger EFNs could have a greater range of variation in their secretion (phenotypic plasticity; Heil, 2015) than smaller EFNs that are structurally constrained by their vascular system (e.g. Villamil *et al.*, 2013). Therefore, the wide variation of EFN traits among plant species impacts the attraction gradient for ant assemblages (Apple and Feener, 2001; Díaz-Castelazo *et al.*, 2005; Baker-Méio and Marquis, 2012), mediating the establishment and outcome of ant–plant mutualism (Díaz-Castelazo *et al.*, 2005; Bronstein *et al.*, 2006; Baker-Méio and Marquis, 2012; Leal *et al.*, 2023).

Plant defence against herbivores has been optimized throughout evolution, reducing the risk of tissue loss so costly to plants (Rhoades, 1979). The optimal defence theory (ODT) holds that higher investments in defence must be allocated to the more vulnerable and valuable tissues (McKey, 1974; Rhoades, 1979; McCall and Fordyce, 2010), thus avoiding or reducing the damage caused by herbivores (Holland *et al.*, 2009). Similar optimization is thought to have occurred for plant biotic defences, such as the function and maintenance of EFNs attended by ants (Calixto *et al.*, 2020). In this case, under the same ontogenetic stage, two fundamental aspects explain when the plant makes nectar available via EFNs to defending ants: (i) the plant's phenology coupled with environmental seasonality (Vilela *et al.*, 2014; Nogueira *et al.*, 2020) and (ii) the existence of control mechanisms for nectar production when the plant phenological stage is favourable for EFN activity (Mondor *et al.*, 2006; Dáttilo *et al.*, 2015). On the one hand, the plant's phenology is more associated with the plant's life history and the variation of extrinsic environmental factors, such as the availability of resources for plants throughout the year (e.g. water and soil nutrient availabilities, Nogueira *et al.*, 2020; Ramos *et al.*, 2022). Conversely, control mechanisms of EFN functioning are more associated with intrinsic plant features reducing the costs of biotic defence or switching them with other defence mechanisms (Moles *et al.*, 2013; Calixto *et al.*, 2015). Plants are expected to secrete more nectar and increase their association with ants when producing new plant tissue under a more favourable abiotic time window (Heil and McKey, 2003; Jones and Koptur, 2015; Nogueira *et al.*, 2020). At this phenological stage, plants are more attractive to ants and, correspondingly, more attractive to herbivores (Coley *et al.*, 2018), essentially because young plant tissues are softer and bear larger amounts of water and nutrients (Coley *et al.*, 2006). However, ant protection benefits depend on the herbivore presence that varies in time and space also due to other ecological factors (Nogueira *et al.*, 2016). Under different herbivory levels, the phenotypic plasticity of EFN functioning plays a central role in determining the cost of these interactions for plants (Heil, 2015). The inducible nature of nectar after herbivory damage (Rudgers and Gardener, 2004; Karban, 2020) may minimize the nectar production cost, attracting more protective ants only when the defence is needed. Therefore, it is likely that protection mutualism is functionally and potentially more beneficial to plants during the development of new leaves or flowers under herbivore attack (Heil and McKey, 2003; Belchior *et al.*, 2016).

The importance of nectar as a resource for the ant community has been confirmed by experimental evidence (Blüthgen and Fiedler, 2004b; Silva *et al.*, 2019; Calixto *et al.*, 2021a). For instance, the availability of extrafloral nectar increases the survival and fitness of ant colonies (Byk and Del-Claro, 2011; Flores-Flores *et al.*, 2018). Nevertheless, ants can display preferences among different sources of nectar available in the environment (Apple and Feener, 2001; Wilder and Eubanks, 2009). More ants are attracted to EFNs that produce greater volumes of nectar (Baker-Méio and Marquis, 2012; Calixto *et al.*, 2020) or even nectar sources with higher sugar concentrations (González-Teuber *et al.*, 2012; Shenoy *et al.*, 2012; Lange *et al.*, 2017). In addition, dominance hierarchy among ant species plays a relevant role in the monopolization of resources (Blüthgen and Fiedler, 2004a). Dominant ant species often take over higher quality resources (Blüthgen and Fiedler, 2004a;

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Fagundes *et al.*, 2017; Flores-Flores *et al.*, 2018) and displace subordinate ant species to lower quality resources (Ribeiro *et al.*, 2019). By monopolizing the best resources, dominant ant species also optimize the protective service offered to plants owing to their higher frequency and aggressiveness against herbivores (Fagundes *et al.*, 2017). Thus, it is expected that plants secreting larger volumes of more concentrated nectar and with greater amounts of sugars represent a more attractive source of resources for ants, especially for dominant ant species, potentially increasing visitation frequency and total abundance of workers on plants.

The legumes are one of the most diverse lineages of plants with a vast morphological variation of EFNs associated with protective ants (Gonzalez and Marazzi, 2018; Marazzi *et al.*, 2019). In the legume genus *Chamaecrista* Moench, despite the single origin of leaf EFNs suggested by Conceição *et al.* (2009), interspecific morphoanatomical differences in EFNs were observed, as well as in the contents of their extrafloral nectar (Baker-Méio and Marquis, 2012; Coutinho and Meira, 2015; Francino *et al.*, 2015). Such variation may create an attraction gradient for ants when these plant species occur in the same area under the same ant and herbivore assemblages. The present study investigated ant–plant protection mutualism in three annual EFN-bearing *Chamaecrista* (*C. duckeana*, *C. calycioides*, and *C. diphylla*) species. Each of the three species bears petiolar EFNs with different sizes and morphologies, representing a gradient of food resource availability for the local ant assemblage. To test this assumption, we initially investigated whether species with larger EFNs offered correspondingly greater nectar volume, concentration, and total mass of sugars to the local ant assemblage. The three species of *Chamaecrista* should also increase nectar production via an induction mechanism after herbivory events, revealing the phenotypic plasticity of EFN functioning. We hypothesize that the induction mechanism would differ across species based on variable EFN sizes. To test this hypothesis, we experimentally investigated the induction mechanisms for nectar secretion after leaf damage under field conditions. We expect species with larger EFNs to respond more intensely to leaf damage by increasing the production of more concentrated nectar with a higher total sugar mass than species with smaller EFNs (see Calixto *et al.*, 2020 for similar predictions across plant organs). Also, plant species with larger EFNs will be more often attended by more aggressive dominant ant species in higher abundance than species with smaller EFNs, since dominant ants mainly access high-quality sources available (González-Teuber *et al.*, 2012; Koptur *et al.*, 2015).

Materials and methods

Study site and focal plant species

The research was carried out in a seasonally dry tropical forest area in the Caatinga domain, Vale do Curu Experimental Farm (3°48′43″S and 39°20′42″W, 80 m asl), municipality of Pentecoste, State of Ceará, Brazil (Supplementary Fig. S1A–D). The local vegetation comprises a dense shrubby Caatinga species growing on sandy loam fluvial neosoils (IPECE, 2009; Araújo *et al.*, 2014). According to Köppen's classification, the climate is semi-arid (BSh), with average annual temperatures ranging from 25 °C to 30 °C and average annual precipitation of ~800 mm with irregular rainfall, usually from February to May (Alvares *et al.*, 2013; Araújo *et al.*, 2014; Pina *et al.*, 2016).

Three species of *Chamaecrista* native to the Brazilian northeastern semi-arid region were selected: *Chamaecrista diphylla* (L.) (Fig. 1I–L); *Chamaecrista duckeana* (P. Bezerra & Afr. Fern.) H.S. Irwin & Barneby (Fig. 1M–R), and *Chamaecrista calycioides* (DC. ex Collad.) Greene (Fig. 1E–H). The three species are subshrubs (~20–60 cm high) with an annual life cycle (therophytes), producing flowers gathered in 1–3 small flowering racemes commonly visited by buzz-pollinating bees (Queiroz and Loiola, 2009; Silva *et al.*, 2014; Flora do Brasil, 2020). *Chamaecrista diphylla* has a wide distribution in the North, Northeast, Center-West, and Southeast geographical regions of Brazil (Flora do Brasil, 2020). This species has a pair of obovate–linear–elliptical leaflets (14–22 × 8–14 mm), long lanceolate stipules (9–25 mm long), and a petiole bearing one or two sessile or slightly stipitate EFNs with a concave secretory surface (Irwin and Barneby, 1982; Queiroz and Loiola, 2009). *Chamaecrista calycioides* occurs in the North, Northeast, Center-West, and Southeast regions of Brazil, has paripinnate leaves composed of 5–12 pairs of oval to lanceolate leaflets (5–13 × 0.7 × 2.3 mm), lanceolate stipules (2–7.5 mm long), and one or two stipitate EFNs on the petiole (Queiroz and Loiola, 2009). *Chamaecrista duckeana* is exclusive to the Northeast region of Brazil (Matos *et al.*, 2019). This species has paripinnate leaves composed of 14–24 pairs of linear–oblong or subobtuse leaflets (8–20 × 3–4 mm), narrowly triangular to lanceolate stipules (8–17 mm long), and a petiole charged with one or two stipitate EFNs (Irwin and Barneby, 1982; Matos *et al.*, 2019).

Differences in EFN sizes among the three *Chamaecrista* species were previously suggested (Irwin and Barneby, 1982), and observations during our field expeditions also pointed out such differences. The size of plant structures is directly associated with aspects of their anatomy and functioning, given that plant size has a scaling effect on plant organs, tissues, and cell types (e.g. Clerx *et al.*, 2020; Fajardo *et al.*, 2020). As plants are modular organisms, scaling relationships are also expected within modules and their structures. In EFNs, previous anatomical studies indicate that larger EFNs in *Chamaecrista* have greater vascularization (Coutinho *et al.*, 2010; Coutinho and Meira, 2015), which could potentially represent a higher production of nectar in these plants. Therefore, EFN size variation may be related to the vascular elements influencing the exudate features and, consequently, the attraction of such nectaries to ants (Villamil *et al.*, 2013). During the field expedition, we noticed that the three species of *Chamaecrista* germinated in the field 2 weeks after the rainy period in March 2020, while flowering began in April and fruiting in May.

Sampling design

Individuals of the three *Chamaecrista* species were systematically chosen and tagged in the field along pre-existing trails in March 2020, totalling 90 individuals (30 plants of each species) spread throughout a total area of 250 000 m^2. Each individual chosen was at least 10 m away from other individuals of the same species to ensure spatial independence between plants and increase the chance that ants visiting each plant belonged to different colonies (Nogueira *et al.*, 2012b). Plant individuals of the three species had equal opportunities of being visited by the local ant assemblage, given that most ant species recorded were observed throughout the entire area. In none of the plants chosen for this study did we detect the presence of hemipteran-secreted honeydew, which could also attract ants (Blüthgen *et al.*, 2006) and bias our results owing to the availability of alternative resources to ants.

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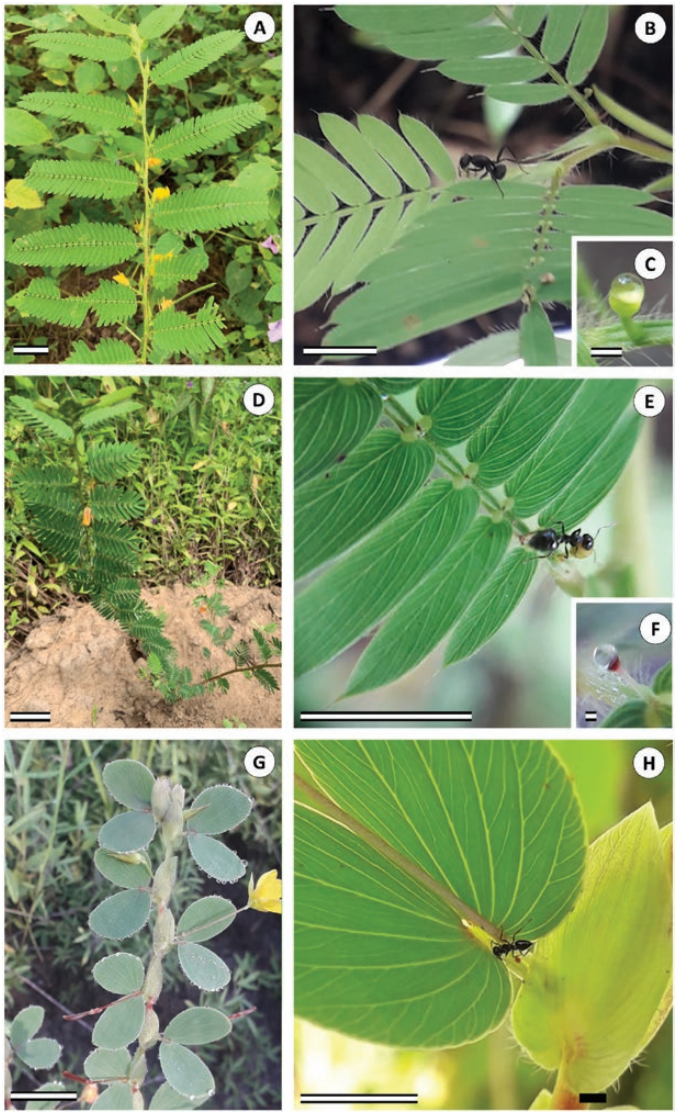


Fig. 1. *Chamaecrista* species used in this study. (A) Subshrub habit of *C. duckeana*. (B and C) *C. duckeana* attended by *Camponotus crassus* feeding on the extrafloral nectaries and a nectar drop on the extrafloral nectary surface. (D) Subshrub habit of *C. calycioides*. (E and F) *C. calycioides* attended by *Crematogaster* sp. 1 feeding on the extrafloral nectaries and a nectar drop on the extrafloral nectary surface. (G) Subshrub habit of *C. diphylla*. (H) *C. diphylla* attended by *Crematogaster* sp. 1. Scale bars: (A), (D), and (G)=20 mm; (B), (E), and (H)=5 mm; (C) and (F)=50 µm.

Considering that the focal species completed their life cycle within the rainy season, all sampling involving EFN collection/fixation, nectar estimation (experimental evaluation), and ant visitation estimation occurred from March to May 2020. Most plants already contained flowers and fruits in June.

Variation in EFN size and extrafloral nectar among Chamaecrista species

Based on the assumption that species with larger EFNs secrete more nectar with higher concentrations and amounts of sugars, we investigated the size of EFNs and nectar characteristics, measuring the volume, concentration, and total mass of sugars in 30 individuals of each *Chamaecrista* species. To quantify the number of leaves with active EFNs, all ants and insects on the leaves were initially removed from each plant at 17.00 h. Afterwards, a non-toxic sticky resin (Tree Tanglefoot® Insect Barrier) was applied over a masking tape attached around the base of the plant stems, which acted as a barrier against new visiting ants. When necessary, branches or leaves from neighbouring plants touching the focal plants were removed, as such branches could allow ants to access the EFNs. The following morning, at ~05.30 h, the number of leaves on the plant and the number of leaves with extrafloral nectar drops in the EFNs, or with at least a glistening at the EFN apex (indicating the presence of nectar), were quantified. This first sampling strategy quantifies the leaves with EFN exudates across the 90 focal plants.

In the next step, we performed measurements of the nectar volume and sugar concentration in 45 of the 90 selected plants in the field (15 individuals of each species). As done previously, plants were cleaned of ants and insects, and the non-toxic sticky resin was reapplied around the base of the plant stem. Then, the plants were bagged with transparent plastic bags that prevented access and consumption of nectar by winged foragers and loss of secretion by rain or dew. As the EFNs in all three species were tiny and exudates accumulated in low volumes under field conditions, bagging the plants allowed the extrafloral nectar to be secreted and accumulated overnight, minimizing nectar losses via evaporation (Nogueira *et al.*, 2015).

The following day, starting at 05.30 h, the accumulated nectar per leaf was collected with 1 µl capillary tubes (Blaubrand® or Drummond Microcaps®). Nectar droplets from all secreting leaves were collected. When the leaves had more than one EFN (maximum of two per leaf), the nectar from both EFNs was considered together. However, most leaves only had a single EFN. A digital caliper (Sylvac IP67 digital Caliper S_Cal EVO) was used to determine nectar volume within the microcapillary. The measurements were then converted into microlitres of nectar. The total concentration of sugars in the nectar was measured using a portable Eclipse low-volume refractometer (Bellingham + Stanley™). The total volume of nectar secreted by all leaves of the same plant was transferred to the prism surface of the refractometer, and the sugar concentration was then read (% Brix). The ambient temperature and relative humidity were also recorded. As the refractometer is calibrated to estimate sugar concentration at 20 °C and the field temperature ranged from 21.6 °C to 32.5 °C during field sampling, sugar concentration values in the nectar were corrected according to the table provided by the refractometer manufacturer (Chinarelli *et al.*, 2021). When the nectar volume produced by all EFNs of a plant was very low (<0.5 µl), it was impossible to estimate that plant's sugar concentration. In such cases, only the nectar volume was registered. From measurements of nectar volume (µl) and sugar concentration (% Brix), we estimated the total sugar mass (mg) produced per leaf using the equation described in Dafni *et al.* (2005).

At the end of fieldwork, one fully expanded leaf (petiole+lamina, without visible signs of damage) per plant from 83 of the 90 initially tagged plants was collected to quantify the EFN size in the laboratory. Initially, leaves were positioned with the abaxial face attached in a Petri dish, and the EFN in the petiole and the adaxial leaf face were observed with a stereomicroscope. We measured the area of the EFN surface (front view) as a proxy of the EFN size. Photographs of EFN surface per plant were taken using a Nikon SMZ 1500 stereomicroscope (Tokyo, Japan) coupled with a Nikon DS-Fi1 digital camera. Then, the EFN area was estimated using the ImageJ software version 1.53e, and values were determined in μm^2 and later converted into mm^2.

Herbivory simulation experiment and induced response in the extrafloral nectar secretion

Next, we investigated the intensification of nectar secretion after herbivory events (phenotypic plasticity) and whether species with the largest

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EFNs had the highest induced response. To accomplish this, an experiment simulating leaf herbivory was carried out on the same 45 plants previously used in the field (i.e. 15 plants for each of the three *Chamaecrista* species). In this case, comparisons were made between the extrafloral nectar secretion already sampled (e.g. under natural field conditions before herbivory simulation) and nectar secretion immediately after herbivory simulation. The first nectar sampling lasted 5 d (nine plants per day). After that, we applied the leaf herbivory on the same subset of plants, following the same sequence previously adopted, and performed the second nectar sampling on plants.

The herbivory simulation was based on previous observations of the herbivory pattern under field conditions in the three *Chamaecrista* species. We considered the average number of leaves damaged by herbivores per plant and the foliar area consumed in the damaged leaves to define the herbivory level in the simulation. In our experiments, the herbivory simulation consisted of removing 20% of the leaf area (the average percentage of herbivory observed previously under natural conditions) with scissors on half of the leaves of each plant. The leaf damage releases volatile jasmonic acid that immediately increases nectar production (Heil, 2015), including in *Chamaecrista* species (Chinarelli *et al.*, 2021). After applying leaf damage at 15.00 h to 19.00 h, the plants were again isolated with non-toxic resin and bagged. The following morning, ~12 h later (05.00 h to 09.00 h), volume and sugar concentrations were measured as described above. We employed the 12 h nectar accumulation protocol following the herbivory simulation experiment applied by Chinarelli *et al.* (2021) that used a legume species model from the same lineage as our study. In our case, we also conditioned the sampling to 12 h (night period), given that during the day the high temperatures under field conditions in the Caatinga domain increase the error associated with nectar measurements, preventing adequate nectar collection even with microcaps.

To compare nectar descriptors before and after herbivory simulation, volume and total sugar mass measurements were standardized to the plant scale (µl per plant and mg per plant). We expected that plants with larger EFNs would have greater control over nectar production when submitted to herbivory; that is, they proportionally increase nectar production in relation to plants with smaller EFNs. This pattern translates into a greater working efficiency of the larger EFNs, given that a greater difference is expected before and after herbivory in relation to those plants whose EFNs are smaller.

Ant visitation to extrafloral nectaries

Ant censuses were carried out in the 90 plants four times in April and May. The first two ant censuses preceded the collection of extrafloral nectar to avoid human interference with ant visitation. This procedure was adopted to prevent plant manipulation (e.g. plant bagging) from interfering with local ant visitation patterns. Two other ant censuses were performed some days after the last nectar measurements. The purpose of these additional observations was to expand our ability to detect ant species assemblages feeding on EFNs. For logistical reasons, we did not observe the nocturnal ant assemblage that is different from the diurnal ant assemblage, although they commonly maintain the same ant genera and dominance hierarchy on plants (Dáttilo *et al.*, 2015; Flores-Flores *et al.*, 2018). In each census, a visual inspection lasting 3 min was performed 1 m away from the plant to quantify the abundance of recruited ants. We never counted the same ant individual more than once in each census. After counting, ants were collected by carefully shaking the plant on a tray containing water. Ants were then transferred to microtubes containing 90% ethanol and stored until species identification in the laboratory. Identifying specimens allowed us to determine ants' diversity (richness and abundance) among the three species.

In the first census, six rounds of observation were carried out on each plant in the morning and three in the afternoon to avoid false absences in detecting ants on each plant. After sampling the set of plants (8–12 plants per day), we returned to the first plant sampled that day and again recorded all data regarding ant visitation. In the second census, carried out 7 d after the first one, a single observation per plant followed by ant collection (90 plants per day) was performed. In the third census, 15 d after the second one, we carried out three observations and one ant collection on each plant in the morning and two in the afternoon (20–25 plants per day). In the fourth and last census, 7 d after the previous one, six observation rounds were carried out on each plant in the morning and three in the afternoon. Ant observation and collection were performed between 06.00 h and 10.00 h, and between 16.00 h and 18.00 h. This procedure was adopted because some species of nectar-foraging ants show a bimodal activity pattern, patrolling in the morning and afternoon at temperatures ranging from 14 °C to 32 °C (O'Dowd, 1979; Oliveira *et al.*, 1999;Vieira *et al.*, 2012; Lima and Antonialli-Junior, 2013).

After identifying the ant species, each was classified into two groups: (i) dominant and (ii) subordinate ant species. For this classification, we used three publications reporting dominant and subordinate ant species or genera in the Cerrado and Caatinga domains (Melati and Leal, 2018; Lange *et al.*, 2019; Calixto *et al.*, 2021b). In these studies, ant dominance was determined according to each ant species' behavioural (aggressiveness) and numerical attributes.

Data analysis

To test the assumption that the three *Chamaecrista* species represent a resource availability gradient for ant assemblage, we initially tested for differences in EFN size and extrafloral nectar descriptors. We constructed five linear models (LMs) in which the superficial area of EFNs (mm^2), the proportion of leaves with active EFNs, nectar volume (ul), sugar concentration (% Brix), and sugar mass of the extrafloral nectar were the response variables, and the species was the categorical factor with three levels. In this case, we performed only one measurement per plant for EFN and extrafloral nectar descriptors. Also, we used Gaussian probability distribution to explain the data variability in these models. After the general analysis, we applied the Tukey *a posteriori* test to evaluate the pairwise difference between plant species.

In the herbivory simulation experiment, we tested whether extrafloral nectar production is enhanced after leaf damage and whether *Chamaecrista* species with larger EFNs have greater control over the secretory activity of EFNs after plant damage. A greater control of larger nectaries occurs if such plants have a higher relative increase in secretion between measurements performed before and after the application of foliar herbivory. We applied three mixed linear models (LMMs) with Gaussian probability distribution to test this expectation, one model for each response variable describing extrafloral nectar: (i) nectar volume per plant; (ii) sugar concentration; and (iii) sugar mass per plant. In these models, the fixed factors were plant species (three levels), herbivory conditions (before and after leaf damage), and the interaction between both factors. As in each focal plant, the nectar descriptors were measured twice before and after the herbivory simulation. The plant identity was included in the models as a random variable. We applied the Bonferroni correction in which the significance level was divided by the number of tests performed ($\alpha=0.05/3=0.016$).

We also extracted the effect size of leaf damage on the extrafloral nectar descriptors by subtracting the nectar values after leaf damage from those before the damage. In this case, values close to zero indicate no effect of leaf damage on nectar production in each *Chamaecrista* species. We built linear models with a Gaussian probability distribution by calculating the leaf damage effect size for each plant. The effect size for each extrafloral nectar descriptor was the dependent variable, and the species was the categorical factor. In this case, when the 95% confidence interval calculated for each species did not reach zero (no damage), we considered the positive effect of the damage on extrafloral nectar induction in the plant species.

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To assess whether ant attendance differs among plant species and if the most attractive plant species monopolize the most protective ant species, we constructed generalized mixed linear models (GLMMs) in which the dependent variables were ant occurrence and number of ants per plant, and the fixed factor was plant species. In the first binomial model, we analysed the ant occurrence data (presence/absence per observation in the plant) to explore whether ant attendance probability differs among the three *Chamaecrista* species. In the second negative binomial model, we investigated whether the number of ants per plant differed among plant species. Here, the response variable is a counting variable and does not have constant variance or normal distribution. Therefore, we used the Poisson and the negative binomial random distribution in the models. Given that the model with the Poisson distribution showed overdispersion of the residuals, we ultimately built a model with the negative binomial probability distribution that no longer revealed overdispersion. Since multiple observations were performed per plant on different days and during different periods, we incorporated the random variables plant identity and period (morning or afternoon) nested within the sampling day in both models. Finally, we investigated whether the maximum number of ant workers per focal plant varied among plant species in a third model. We used the same structure as the previous models, but with modifications. We did not include random factors for this analysis since the maximum number of ants in each plant was a single observation in time. Therefore, in this test, we did not have repeated measurements of ants per plant.

Subsequently, we investigated the visitation pattern of the dominant ant species in the three *Chamaecrista* species, given that such ants are potentially more protective against herbivores. For this, we initially categorized ant species attending plants into dominant and subordinate species (see criteria above based on the literature). Two dominant ant species were identified (*Camponotus crassus* and *Pheidole* sp.1), and we built two GLMMs with a negative binomial random distribution. For each dominant species, the dependent variable was the number of ants per plant, and the categorical factor was plant species. We used the same random factors as those in the models described above. In all cases, *a posteriori* Tukey tests were applied to detect the pairwise differences among plant species.

The assumptions of normality and homogeneity of residual variances were verified in the Gaussian models. We performed a graphical visual inspection of the residuals for assumption evaluation in those cases. Additionally, we evaluated the degree of overdispersion of the residuals in the generalized models. For the execution of all statistical analysis and graphing, we used RStudio software, version 1.3.959 (R Core Team, 2021), and the following additional packages: glmmTMB (Bolker, 2021), glmmADMB (Skaug *et al.*, 2018), ggeffects (Lüdecke, 2021), emmeans (Lenth *et al.*, 2021), MASS (Brian *et al.*, 2021), and ggplot2 (Wickham *et al.*, 2021).

Results

Variation of extrafloral nectary size and nectar production across Chamaecrista *species*

The surface area of EFNs differed among the *Chamaecrista* species (Table 1; Fig. 2A). *Chamaecrista duckeana* plants have EFNs with the largest size, while the EFNs of *C. calycioides* and *C. diphylla* plants have the smallest area (Table 1). The EFNs of *C. duckeana* had an average area of 0.46 ± 0.05 mm^2 (mean ±SE). It was about four times larger than the EFN area of *C. calycioides* and seven times larger than the EFN area of *C. diphylla* (Fig. 2A).

On average, all three species of *Chamaecrista* had 30% of their leaves secreting extrafloral nectar (Table 1; Supplementary Fig. S2). In contrast, the extrafloral nectar volume (µl per leaf), sugar concentration (% Brix), and sugar mass (mg per leaf) produced on average by each EFN differed among the *Chamaecrista* species (Table 1). The nectar volume secreted by the EFNs of *C. duckeana* was 0.05 ± 0.03 µl per leaf, about three times higher than the volume secreted by the EFNs of *C. calycioides* and 12 times higher than that of *C. diphylla* EFNs (Fig. 2B). The nectar sugar concentration (% Brix) was similar between *C. calycioides* and *C. duckeana* plants, ~4.1 ± 0.1%. Both species had nectar sugar concentration twice as high as that of *C. diphylla* plants (Fig. 2C). The nectar sugar mass (mg per leaf) of *C. duckeana* was 0.003 ± 0.007 mg, about three times higher than the extrafloral nectar of *C. calycioides* and 10 times higher than that of *C. diphylla* (Fig. 2D).

Herbivory induces distinct responses in extrafloral nectar production among Chamaecrista *species*

The effect of leaf damage on nectar production depended on the *Chamaecrista* species (herbivory×species interaction: $F_{2,42}$=3.8; P=0.029). While *C. duckeana* plants secreted on average 1.05 µl more nectar after leaf damage (Fig. 3A), *C. calycioides* secreted on average 0.62 µl more nectar after leaf damage than before (Fig. 3A). On the other hand, *C. diphylla* plants, with the smallest EFN size, secreted a similar amount of nectar,

Table 1. Variation of extrafloral nectary (EFN) size and extrafloral nectar production among the three *Chamaecrista* species under field conditions

EFN descriptors	C. diphylla		C. calycioides		C. duckeana		Statistics			Figures
	Mean	SE	Mean	SE	Mean	SE	*F*	*P*	N	
Secretory area (mm^2)	0.07	0.05	0.11	0.06	0.46	0.06	335.1	<0.001	83	Fig. 2A
Nectar volume (µl per leaf)	0.005	0.007	0.02	0.01	0.06	0.02	6.01	0.005	45	Fig. 2B
Sugar concentration (% Brix)	2.21	0.02	4.20	0.70	4.14	0.42	3.99	0.03	45	Fig. 2C
Sugar mass (mg per leaf)	0.0003	0.0002	0.001	0.0007	0.004	0.001	9.04	0.001	45	Fig. 2D
Proportion of leaves with active EFNs (plant)	0.30	0.06	0.31	0.06	0.29	0.04	0.04	0.96	90	Supplementary Fig. S2
Nectar volume (µl per plant)	0.07	0.10	0.32	0.37	0.63	0.68	7.15	0.002	45	–
Sugar mass (mg per plant)	0.004	0.005	0.001	0.02	0.003	0.04	5.12	0.01	45	–

The EFN area corresponds to the superficial gland area (top view).

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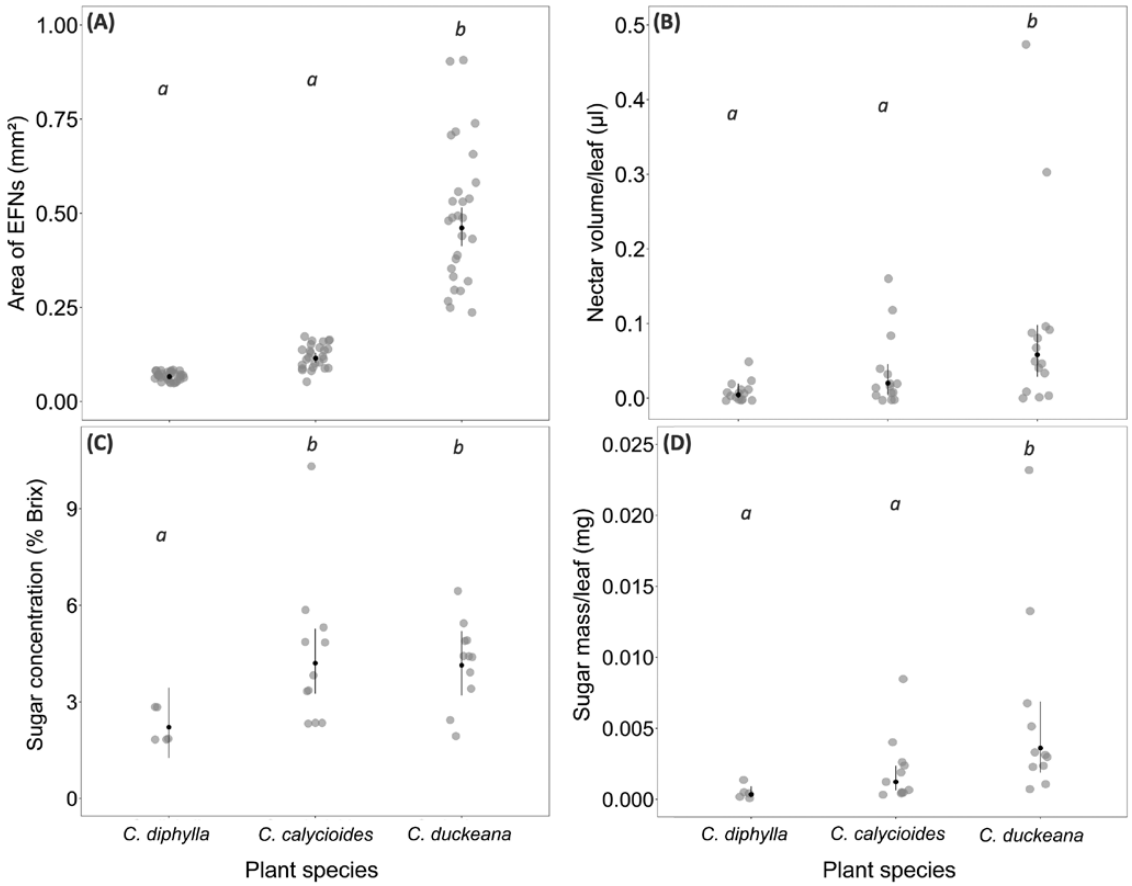


Fig. 2. Size variation of extrafloral nectaries and nectar descriptors among the three *Chamaecrista* species under field conditions. (A) Surface area of each extrafloral nectary (mm^2). (B) Extrafloral nectar volume (µl per leaf). (C) Sugar concentration (% Brix). (D) Sugar mass (mg per leaf). The midpoint represents the mean, the vertical line represents the 95% confidence interval, and the grey dots are the raw data. Different letters represent differences among the studied *Chamaecrista* species ($P<0.05$).

~0.06 µl, both before and after herbivory (Fig. 3A). In the complementary analysis, the effect size of leaf damage was positive and higher in *C. duckeana*, followed by *C. calycioides* and, finally, no difference in *C. diphylla* (Supplementary Fig. S3A).

Sugar concentration in the extrafloral nectar increased slightly after applying leaf damage ($F_{1,21}=4.8$; $P=0.04$) and also differed among *Chamaecrista* species ($F_{2,21}=6.4$; $P=0.007$). However, both factors did not interact ($F_{2,21}=0.05$; $P=0.95$). On average, after applying leaf damage, *C. duckeana* and *C. calycioides* plants increased their sugar concentration to 5.6 ± 0.5% and 5.3 ± 0.4%, respectively (Fig. 3B), while *C. diphylla* increased to 2.8 ± 0.9 % (Fig. 3B). Although sugar concentration in *C. diphylla* increased after damage, we point out that in *C. diphylla*, sugar concentration was estimated based only on three individual plants. The low sample size was due to the reduced volume of nectar secreted in most plants before the application of leaf damage, preventing the nectar concentration estimations in most cases. Thus, our estimates of the impact of leaf damage on nectar sugar concentration for *C. diphylla* are less reliable than estimates for the other two species. In the complementary analysis, leaf damage effect size was not different in the three species (Supplementary Fig. S3B).

Sugar mass production increased after the application of leaf damage ($F_{1,21}=31.1$; $P<0.0001$) and differed among the *Chamaecrista* species ($F_{2,21}=21.3$; $P<0.0001$). However, no interaction occurred between herbivory and plant species ($F_{2,21}=2.9$; $P=0.073$). Although we did not detect an interaction between them, the complementary analysis revealed that the leaf damage effect size was positive and higher in *C. duckeana*, followed by *C. calycioides*, and, finally, not different in *C. diphylla* (Supplementary Fig. S3C). Plants of *C. duckeana* increased sugar mass on average by 0.10 mg per plant after herbivory (Fig. 3C), while *C. calycioides* increased by 0.04 mg per plant (Fig. 3C). *Chamaecrista diphylla* plants did not increase sugar mass production after herbivory (Fig. 3C).

Ant assemblage and numerical dominance pattern of species

At the end of 1365 observations on 90 focal plants across the three *Chamaecrista* species (~15 observations per plant),

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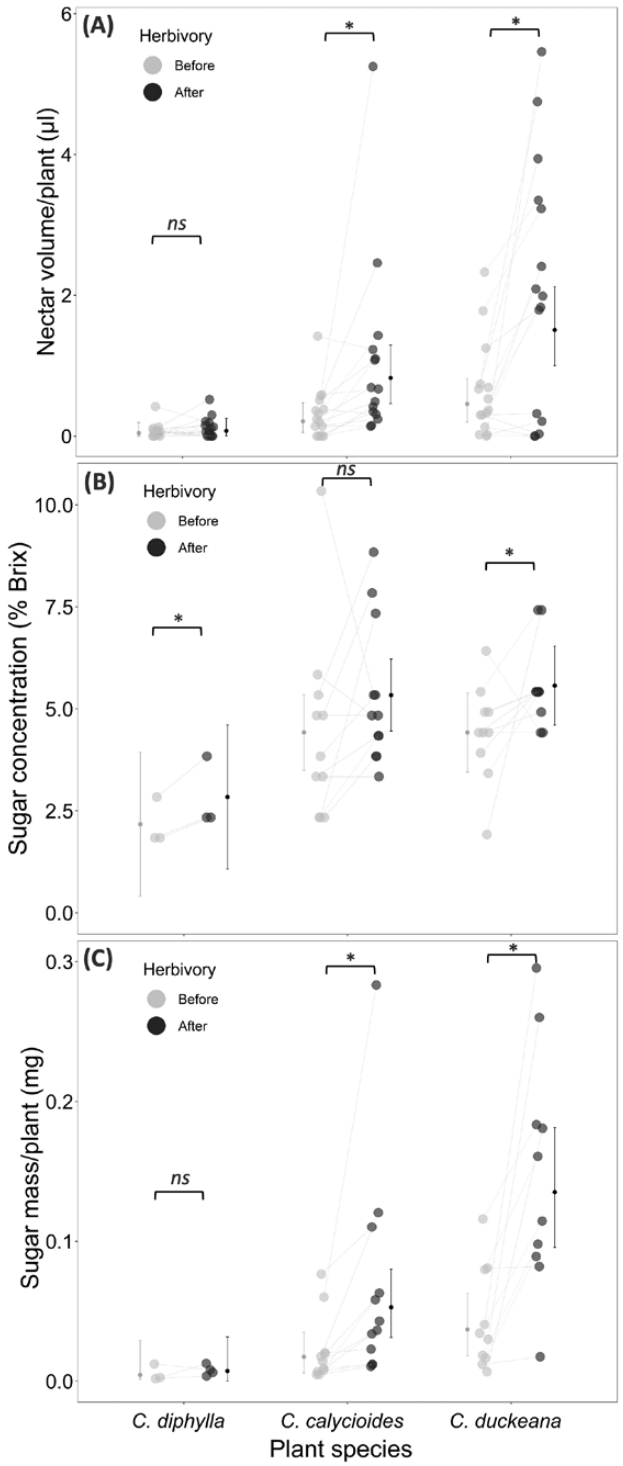


Fig. 3. Extrafloral nectar production before and after leaf damage in the three *Chamaecrista* species. The experimental application of leaf herbivory represented a removal of 20% of the leaf area in 50% of leaves in each plant. (A) Extrafloral nectar volume (µl per plant). (B) Sugar concentration (% Brix). (C) Total sugar mass (mg per plant). The midpoints represent the means, the vertical lines represent the 95% confidence intervals, and the grey dots are the raw data. We highlight in the graphs the paired comparisons before and after the application of leaf damage: ns=not significant; *significant with $P<0.016$ ($\alpha=0.05/3$ with Bonferroni correction).

we detected 191 worker ants visiting the EFNs. These ants belonged to six genera and nine species (Supplementary Table S1). Two of the nine ant species—*Camponotus crassus* and *Pheidole* sp. 1—were classified as dominant ant species based on the literature (see the Materials and methods). Both ant species had a higher relative frequency (Supplementary Fig. S4A) and aggressiveness in our focal plants. Also, the two dominant ant species were mainly recorded in *C. duckeana* plants Supplementary Fig. S4D).

Ant visitation pattern on extrafloral nectaries of the three Chamaecrista *species*

The probability of ant attendance on EFNs was different among the *Chamaecrista* species ($F_{2,87}=18.5$; $P<0.001$; Fig. 4). Plants of *C. duckeana* were three times more likely to be attended by ants (Fig. 4). EFNs were visited by eight ant species, including two dominant ant species, *C. crassus* and *Pheidole* sp. 1. On the other hand, *C. calycioides* and *C. diphylla* had a very low probability of being attended by ants (Fig. 4) and were visited by six and three ant species, respectively. The dominant ant species rarely visited *C. calycioides* and *C. diphylla*. *Crematogaster* sp. 1 was the most abundant ant species attending plants, but it was observed to feed more frequently on EFNs of *C. calycioides* and *C. diphylla*.

The average number of ants observed on plants during the censuses differed among *Chamaecrista* species ($F_{2,1246}=21.3$; $P<0.001$). *Chamaecrista duckeana* had three times more worker ants visiting EFNs than the other two plant species (Fig. 5A). *Chamaecrista calycioides* and *C. diphylla* did not differ in the number of ants visiting EFNs (Fig. 5A). Similarly, the maximum number of ants per plant was around twice higher in *C. duckeana* than in the other two species, which did not differ from each other ($F_{2,87}=5.70$; $P=0.004$; Fig. 5B).

The visitation pattern of the two dominant ant species differed among the three *Chamaecrista* species (Fig. 5C–F). The ant number of *C. crassus* differed among plant species ($F_{2,1247}=17.5$; $P<0.001$), being higher in *C. duckeana* plants (Fig. 5C). Similarly, the maximum number of *C. crassus* was more elevated in *C. duckeana* ($F_{2,87}=12.6$; $P<0.001$; Fig. 5D). The ant number of *Pheidole* sp.1 also differed among the species of *Chamaecrista* ($F_{2,1247}=38.4$; $P<0.001$), again being higher in *C. duckeana* (Fig. 5E). Similarly, the maximum number of *Pheidole* sp.1 was higher in *C. duckeana* compared with the other two species ($F_{2,87}=52.1$; $P<0.001$; Fig. 5F).

Discussion

Our study investigated the regulation of ant–plant mutualism in three sympatric *Chamaecrista* species bearing different EFN sizes in the Caatinga domain of northeastern Brazil. We hypothesized that the species with larger EFNs would produce more extrafloral nectar by volume, concentration, and

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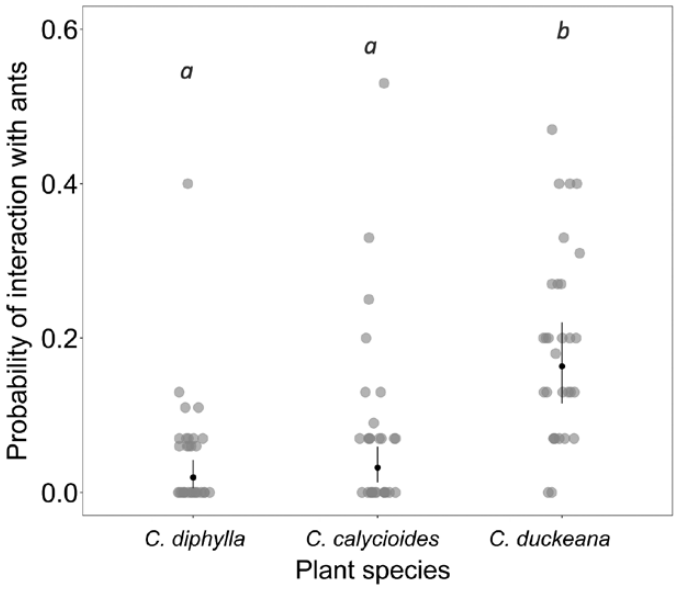


Fig. 4. Probability of interactions with ants among the three *Chamaecrista* species considering the set of observations per plant. The midpoint represents the mean, the vertical line represents the 95% confidence interval, and the grey dots are the raw data. Different letters represent differences between the *Chamaecrista* species ($P<0.05$).

sugar mass, and that such species with larger EFNs would have the greatest induced response after herbivory compared with the other two species. *Chamaecrista duckeana* corroborated our case in both instances, revealing the highest phenotypic plasticity of EFN functioning. Indeed, the inducible response of *C. duckeana* after herbivory evidenced a greater control of its EFN secretory activity over time. Moreover, *C. duckeana* was the most attractive plant species, interacting with a larger number of dominant ant species often more protective against herbivores. Conversely, the other two *Chamaecrista* species produced smaller amounts of nectar and sugar mass, showed a less evident induced response, and interacted less with ants, seldom with dominant ant species. These aspects reinforce the importance of the EFN size in determining its functionality and the phenotypic plasticity of nectar production.

The EFN size of the three *Chamaecrista* species was an excellent indicator of EFN productivity (Fig. 2), directly impacting the ant assemblage attending plants. Although this pattern has already been reported in the literature (Apple and Feener, 2001; Rudgers, 2004; Baker-Méio and Marquis, 2012), it has not always been verified. This happens because plant attractiveness is determined by the size of the secretory structure and the number of structures concurrently secreting nectar in the whole plant (Nogueira *et al.*, 2020). Thus, in addition to EFN size, the number of leaves with active EFNs may vary among individual plants and species since plants may be at different phenological stages in the field (Belchior *et al.*, 2016; Nogueira *et al.*, 2020). In our study, the three *Chamaecrista* species had a similar number of newly expanded leaves secreting extrafloral nectar (~30% of the leaves). Therefore, all three species were at the same phenological stage during the study. This phenological pattern of EFN activity is explained because all studied species are annual subshrubs at the same growth stage, with leaf sprouting and extrafloral nectar production synchrony during the rainy season. In the Caatinga domain, seasonality is well marked, and the rainy season is short (da Silva *et al.*, 2017). Synchronism in plant phenology of most species inhabiting such domain is favoured (Borchert, 1996). This pattern is very different from what is observed in other systems where seasonality is less pronounced, and differences in the phenological pattern across plant species (e.g. differential production of leaves) are more critical for the establishment of ant–plant interactions than the EFN secretory area (Nogueira *et al.*, 2020).

Interspecific variation of herbivory-induced response in extrafloral nectar secretion

Although the studied species had different responses, our results showed a marked induced response in extrafloral nectar secretion after leaf damage (Fig. 3). Most plants subjected to simulated leaf herbivory increased EFN productivity in at least one extrafloral nectar descriptor, corroborating the phenotypic plasticity hypothesis. The larger the EFN size, the greater the intensity of the plant's response in the volume and sugar mass of the extrafloral nectar secretion. The moderate increase in nectar sugar concentration (% Brix) after leaf damage was similar for the three species. Increases in extrafloral nectar production induced by leaf herbivory events are consistent with the ODT (McKey, 1974; Rhoades, 1979). The ODT predicts that induced defences are favoured over constitutive defences if defences are costly and sporadically unnecessary (Karban, 2020). Consistent with this prediction, plants can minimize defence investment costs through the induced production of extrafloral nectar at specific times (Holland *et al.*, 2010; Bixenmann *et al.*, 2011). EFN activity is modulated according to the ecological context in response to different factors, such as herbivore attacks and soil water availability (Rios *et al.*, 2008; Rocha *et al.*, 2019). During the rainy season, the studied species had no water restriction. Nectar secretion dynamics and EFN productivity, especially in *C. duckeana* and *C. calycioides*, were experimentally responsive to herbivory. In this system, herbivory is typically sporadic and unpredictable, although intense from time to time (Oliveira *et al.*, 2012). This pattern of activity of herbivores on plants may explain mechanisms for increasing EFN productivity only when damage is inflicted on more vulnerable newly produced plant tissues. Such a control mechanism may have been optimized over evolutionary time, maximizing the benefits and reducing costs of ant–plant–herbivore interaction (Karban, 2020).

The highest increase in all nectar descriptors in *C. duckeana* plants with larger EFNs demonstrates that this species more efficiently controls extrafloral nectar secretion. This pattern of EFN function enables *C. duckeana* to provide a more

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significant net benefit to plants engaged in ant–plant–herbivore interactions over time, reducing secretion costs in the absence of herbivores. Therefore, species with larger EFNs optimize plant resources for biotic defence and increase secretion only when protection is needed. Although the induced response on EFN function has been demonstrated for other plant species (Holland *et al.*, 2009; Calixto *et al.*, 2020; Chinarelli *et al.*, 2021), variation in the intensity of the herbivory effect according to the EFN size reveals a distinct control mechanism across plant species that could be a consequence of the association between EFN size and morphology. Indeed, nectar secretion rates are determined by the proportion of secretory area in the epidermis, the amounts of nectariferous tissue, and its degree of vascularization (Díaz-Castelazo *et al.*, 2005; Nicolson *et al.*, 2007). Therefore, in addition to secreting more significant nectar amounts, larger EFNs, such as those in *C. duckeana*, also have greater control over secretion. Additionally, knowing that EFN size and morphology vary between closely related species (Apple and Feener, 2001; Nogueira *et al.*, 2012a; Coutinho and Meira, 2015; Marazzi *et al.*, 2019), or even within the same species (Rios *et al.*, 2008; Nogueira *et al.*, 2015), these attributes are fundamental for the very understanding of interactions more or less costly in the community.

In contrast to *C. duckeana*'s responses to leaf damage, *C. diphylla* with the smallest EFN did not express a damage-induced response in extrafloral nectar volume and sugar mass. An alternative explanation for the differential response of species to leaf herbivory could be related to the level of damage inflicted. It is known that the magnitude of the induced response also depends on the type and level of damage imposed on the plants (Pulice and Packer, 2008; Calixto *et al.*, 2020). Therefore, *C. diphylla* individuals may be induced only after more intense herbivory events or even use other defence strategies. Field observations did not show higher natural herbivory levels for *C. diphylla* compared with the other two studied species, which means that the same herbivory intensity was applied to all three species. Therefore, the absence of a whole induction mechanism on nectar secretion could explain the lack of induced response for *C. diphylla*. Furthermore, the induced response mechanism is expected to be favoured if the production of the defence is costly, at least at some point in the plant's life (Karban, 2020). If nectar production is minimal over time and, hence, probably inexpensive, as seems to be the case in *C. diphylla*, then there is a lower probability for the evolution of control mechanisms for reducing the nectar secretion costs. Given the reduced number of studies that have experimentally tested induced response in nectar secretion after herbivory (e.g. Agrawal and Rutter, 1998; Heil *et al.*, 2001; Ness, 2003; Calixto *et al.*, 2020), especially in the field, the differences we found here among the three species demonstrate that we still know very little about the variation of these mechanisms among species (see Chinarelli *et al.*, 2021) and that any generalization is hasty pending future comprehensive studies.

Although extrafloral nectar production in the studied species is lower in the absence of herbivory, generating a lower cost for plants (e.g. Katayama and Suzuki, 2011), a gap still exists in the direct quantification of extrafloral nectar costs. In contrast, mechanisms for inducing secretion in plants under herbivore attack are evidence of a synchronized system that increases investment in extrafloral nectar production, but only when the benefits are more likely to outweigh the costs for plants (Karban, 2020). Therefore, the response induced after herbivory is taken as indirect evidence of EFN secretion cost. In *C. fasciculata*, for example, natural selection favours plants that produce more nectar in the presence of ants and herbivores (Rutter and Rausher, 2004). The production of extrafloral nectar in the absence of ants and herbivores imposes costs on these plants (Rutter and Rausher, 2004), especially in the absence of other natural enemies of herbivores on plants (e.g. wasps and spiders; Del-Claro *et al.*, 2016). Therefore, selection should favour extrafloral nectar production in populations where ant protection services have higher relative value to plants, with the benefits of plant defence production exceeding its costs (Nogueira et al., 2015, 2016).

Ant diversity and visitation to the extrafloral nectaries of Chamaecrista

Ant visitation patterns varied by *Chamaecrista* species (Figs 4, 5). A greater diversity was associated with *C. duckeana*, the EFNs of which are larger and more productive. On the other hand, *C. calycioides* and *C. diphylla*, the EFNs of which are smaller and less productive, supported a gradually lower diversity. In this case, the three plant species constitute a gradient of ant attraction at the study site, and this is essentially due to the positive relationship between EFN size and productivity, which has also been demonstrated in other studies (Rios *et al.*, 2008; Baker-Méio and Marquis, 2012). Such studies showed that greater ant density and richness were associated with plant populations having the greatest number of secreting EFNs and the most concentrated nectar (Rios *et al.*, 2008; Baker-Méio and Marquis, 2012; Nogueira *et al.*, 2015). Therefore, our results confirm that both EFN and nectar attributes determine different ant visitation patterns across the three plant species due to differences in nutritional requirements or the resource's dominance capacity of ants.

Extrafloral nectar is a primary food resource for many ant species (Davidson, 1998; Blüthgen *et al.*, 2004; Heil, 2015), contributing to increases in colony growth, survival, and fitness (Byk and Del-Claro, 2011; Calixto *et al.*, 2021a), spatial structuring of assemblages (Silva *et al.*, 2019), and, ultimately, the ecological domain of ants (Kay *et al.*, 2014). As a rule, increasing nectar availability increases ant abundance and richness in plants (González-Teuber *et al.*, 2012; Belchior *et al.*, 2016; Lange *et al.*, 2017) and could also change species composition (Ribeiro *et al.*, 2018). The highest EFN productivity in *C. duckeana* conditioned a higher probability of ant attendance,

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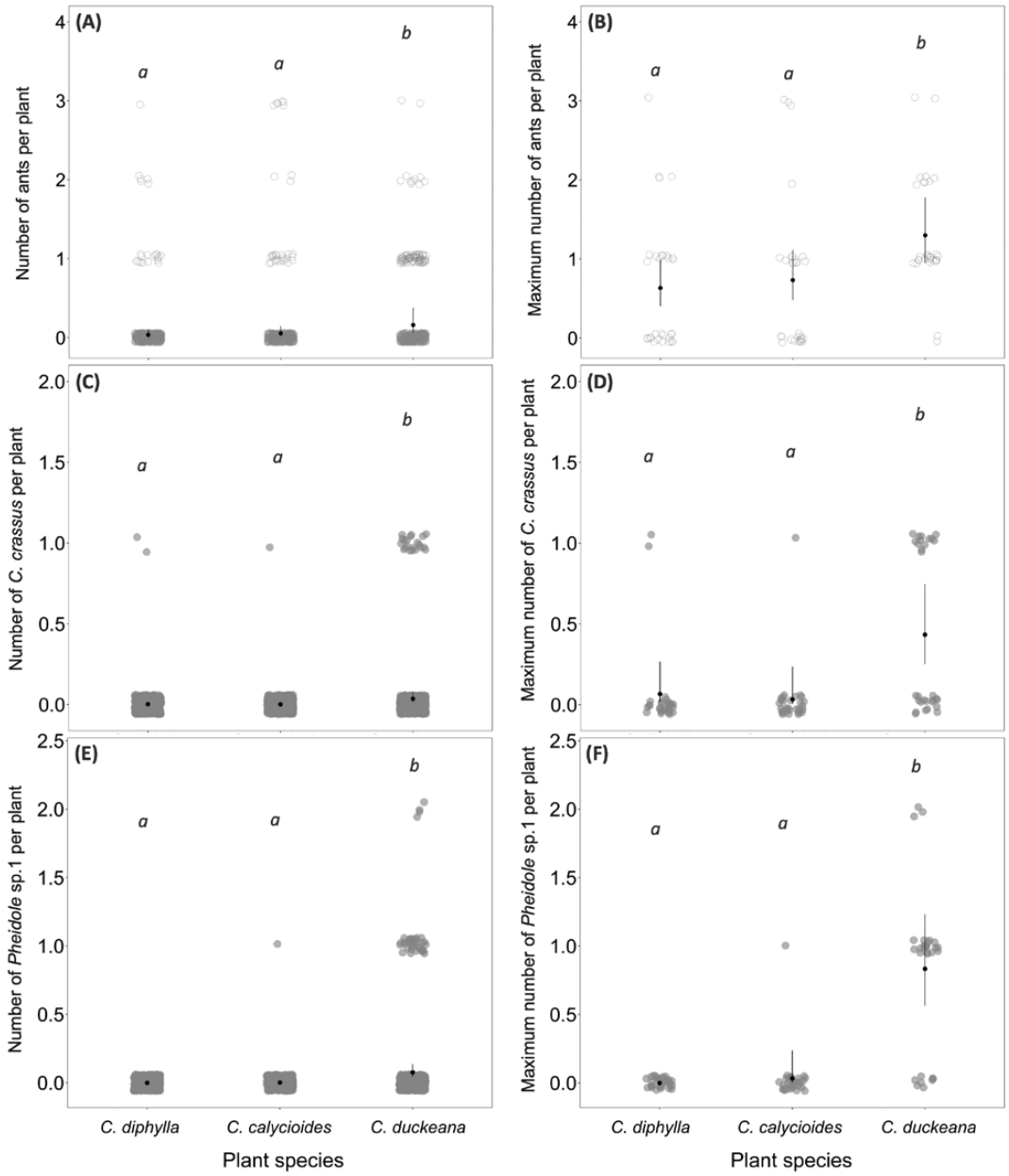


Fig. 5. Ant visitation pattern among the three *Chamaecrista* species. (A) Considering all observations, the number of ants per plant and (B) the maximum number of ants per plant. (C) Number of ant workers of the dominant ant species *Camponotus crassus* per plant considering all observations and (D) the maximum number of *C. crassus* per plant. (E) Number of ant workers of the dominant ant species *Pheidole* sp. 1 per plant considering all observations and (F) the maximum number of *Pheidole* sp. 1 per plant. The midpoint represents the mean, the vertical line represents the 95% confidence interval, and the grey dots are the raw data. Different letters represent differences between the *Chamaecrista* species ($P<0.05$).

abundance, and species richness (eight out of the nine species recorded). Plants with more productive EFNs usually experience greater ant diversity and recruitment (Bixenmann *et al.*, 2011; Fagundes *et al.*, 2017; Pacelhe *et al.*, 2019), including in the Caatinga (Souza *et al.*, 2022). Also, the interaction may not only depend on the plant aspects. The spatial distribution of ant nests (Moura *et al.*, 2022) and even the indirect effects between EFN-bearing plants via shared ants (Moura and Del-Claro, 2023) could contribute to differences across plant species. Since plants in our study occur in the same area, with a minimum distance between them, and are subject to the same environmental conditions, ant diversity and interaction probability across the three *Chamaecrista* species mainly result from variations in EFN productivity.

In the context of protection mutualisms, one of the main advantages of widespread interaction with multiple potential partners is that different ant species have unique behaviours in their interactions with herbivores (Miller, 2007; Heil, 2008).

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For example, interaction with dominant ant species would be more advantageous for plants, as these species tend to be more aggressive than subordinate ones (Andersen, 1992). In the Caatinga, the *Camponotus* and *Pheidole* ants are considered more aggressive towards other ants and thus considered behaviourally dominant (Melati and Leal, 2018). Also, species of both genera are frequent and numerically dominant ants (Melati and Leal, 2018). In fact, numerical dominance usually correlates with behavioural dominance in ant species (Davidson, 1997; Blüthgen and Fiedler, 2004a). In our study, two ant species were considered dominant (*C. crassus* and *Pheidole* sp.1) and visited almost exclusively the EFNs on *C. duckeana* plants. Plants that provide better quality nectar concentrate more interactions with dominant ant species, reducing herbivory and the frequency of other ant species (González-Teuber *et al.*, 2012; Koptur *et al.*, 2015). Therefore, *C. duckeana* probably has its protection enhanced by interaction with dominant ant species more often, while *C. diphylla* and *C. calycioides*, which interacted more frequently with subordinate ant species, are probably less protected.

While the ants most frequently feeding on EFNs of *C. duckeana* were the two dominant ant species, plants of *C. diphylla* and *C. calycioides* were mainly visited by subordinate ants. The larger EFNs in *C. duckeana* provide a greater reward per plant unit for dominant ant species than the smaller EFNs of *C. calycioides* and *C. diphylla*, which are not attractive enough for dominant ant species. A similar pattern was observed among populations of *Anemopaegma album* Mart. ex DC. (Bignoniaceae), where *Camponotus* species dominated populations, the plants of which have greater EFN abundance per leaf, while the population with lower abundance was dominated by the ant *Crematogaster crinosa* (Nogueira *et al.*, 2015). In our study, *Crematogaster* sp. 1 is probably displaced to less attractive plants, while *C. crassus* and *Pheidole* sp. 1 dominate the best nectar sources. Thus, dominant ant species may competitively exclude subordinate ant species, conditioning their foraging behaviour on lower quality sources. In our case, the two dominant ant species were never found together or with subordinate ant species foraging on the same plant, suggesting competitive exclusion generated by locally dominant ant species (Andersen, 1992; Parr *et al.*, 2005). In these cases, the spatial partition of local resources favours the co-existence of a diversity of ant species in the system (Coley *et al.*, 2018).

Conclusion

Our study demonstrates that congeneric plant species whose populations share the same habitat and local ant assemblage differ in EFN size and productivity and, consequently, in the probability of ant attendance, abundance, and richness of ants feeding on EFNs. We also identified that differential EFN size is an essential attribute for determining the intensity of the damage-induced response of EFNs across plant species; that is, the phenotypic plasticity of nectar production can be predicted by the secretory structure size. Our results demonstrate that differences in extrafloral nectar quantity and quality among *Chamaecrista* species generate a gradient of attraction to ants, shaping ant diversity and dominance patterns. Finally, the differences in EFN function detected here necessarily impose constraints on the benefits that biotic defence can generate for plant species. Consequently, in the presence of cost-reducing mechanisms (e.g. inducible EFN response under herbivory), plant species with larger EFNs have a higher chance of functioning as an effective defence against herbivores over time.

Supplementary data

The following supplementary data are available at *JXB* online.

Table S1. Ant species feeding on extrafloral nectaries of *Chamaecrista* species.

Fig. S1. Study site.

Fig. S2. The proportion of leaves with active extrafloral nectaries (EFNs) secreting nectar among *Chamaecrista* species.

Fig. S3. Effect sizes of leaf damage applied to plant leaves on extrafloral nectar descriptors in *Chamaecrista* species.

Fig. S4. Relationship between the relative frequency (%) and the average number of ants per plant across ant species.

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Authors contributions

CLA, AN, and IACC: conceptualization and experimental design; CLA: data collection; CLA and AN: data analysis; REV: identification of the ants; AN and IACC: funding acquisition; CLA: leading the writing. All authors contributed to the writing and gave final approval for publication.

Conflict of interest

The authors have no conflict of interest to declare.

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Data availability

Data of nectar secretion descriptors before and after leaf damage and ant–plant interactions are available from the Dryad Digital Repository: doi:10.5061/dryad.547d7wmbw (Alencar *et al.*, 2023)

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